

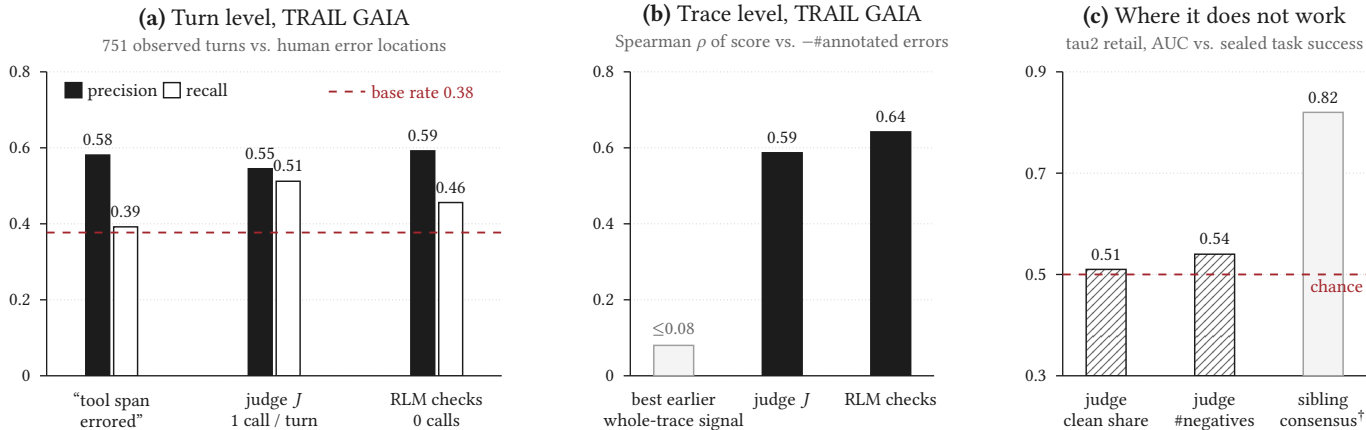


Figure 3. Results. (a) On TRAIL GAIA, the judge’s -1 turns and the checks it was distilled into land on human-annotated errors at well above the base rate. The checks match the judge’s precision without a model call. A plain “tool span errored” rule is nearly as precise but finds fewer errors (recall 0.39 vs. 0.46). (b) At trace level, reading reactions carries signal that reading whole trajectories did not: every earlier trajectory-shape signal on this split reached $\rho \leq 0.08$. (c) On tau2 retail the judge is at chance, because tau2’s simulated user never pushes back and its failures live in a goal state no reaction shows. [†]Sibling consensus (full 456-trace set) needs $k > 1$ rollouts per task; the judge bars are a 120-trace subset. On TRAIL SWE (31 traces, not shown), a Spearman carries roughly ± 0.35 of noise and two proposal runs disagree (checks $\rho = 0.43$ vs. 0.15 against *overall*), so nothing there separates the scorers. Model: nemotron-lightning-3p5-30b-a3b, one vote at temperature 0. Full tables are in docs/next-state-verifier.md.